

Where Do Semantics Live? A Homonym-Based Study of Semantic Disambiguation in Large Language Models

Julia Hagen, 22430997

Friedrich-Alexander-Universität Erlangen-Nürnberg

Abstract—Layer-wise comparisons offer one way to test whether artificial and biological systems organise similar functions across processing stages. We apply this approach to contextual word meaning in transformers. The study uses seven two-sense homonyms in four bidirectional encoders and four causal decoders. It separates three dimensions: layer depth, token position, and incrementally available context. Sense-conditioned representations are evaluated with held-out nearest-centroid margins and the Generalized Discrimination Value (GDV).

Contextual sense was widely decodable, but its strongest readout varied across depth. Profiling-selected layers reached 0.917 held-out adequacy, compared with 0.893 at the final layer. The crossed model-word interval included zero, and effects varied across models and homonyms. Adequacy-based cross-word selection reached 0.914 adequacy, compared with 0.892 for GDV-based selection. The two criteria also selected different depths in most folds. Global class separation and item-level decodability therefore captured different properties of the activation geometry.

Context-position analyses showed that semantic readout also depends on causal context availability. In causal decoders, homonym-position adequacy was 0.500 when the resolver followed the homonym. At the sentence-final period, after the resolver had become available, adequacy reached 0.761. Among observations that were inadequate at the homonym, 67.9% became locally decodable at the endpoint.

Incremental context-conflict experiments showed broad updating and persistent context effects. Resolving information moved a fixed sentinel toward the authored sense in 77.2% of observations. Among initially primed states, 40.3% crossed into the resolved-sense region. Conflict endpoints remained 0.140 below matched non-conflicting endpoints containing the same homonym and resolver.

Contextual sense was distributed across depth and depended on the readout criterion, token position, available context, and context history. These results motivate functional comparisons between humans and language models at the level of prediction, integration, conflict, and revision.

Index Terms—large language models, semantic representations, homonyms, contextual updating, representational geometry, layer selection, encoders, causal decoders

I. INTRODUCTION

Comparisons between artificial and biological information-processing systems naturally focus on functional analogues: whether both systems perform similar computations and how those computations are organised internally. Visual neuroscience provides an established example. Brain-Score evaluates artificial vision models against neural and behavioural benchmarks and relates model layers to cortical targets along the primate ventral visual pathway [1]. This stage-wise comparison

motivates an analogous question for language: do transformer layers organise linguistic functions in a reproducible order?

Human comprehension transforms perceptual input into lexical, syntactic, and semantic representations that support action and language production. These functions are partly ordered and interact through prediction, bottom-up evidence, and top-down context [2]. Transformers also transform input through a sequence of internal representations before producing an output. Their depth may therefore organise linguistic functions across successive processing stages.

Previous probing studies have found that linguistic information is distributed non-uniformly across transformer depth [3]. Different layers can support different forms of linguistic analysis. The present study tests one restricted part of this proposed ordering: whether contextual word sense reaches its most reliable and reproducible readout within a particular depth region.

The present study tests a restricted version of that hypothesis: whether contextual word sense becomes most clearly and reproducibly expressed within a particular region of transformer depth. This question concerns layer-level population geometry rather than individual interpretable features. Conceptual features may be distributed or superposed across many units and layers [4–6], while their combined expression could nevertheless become maximally decodable at a narrower stage. Whether such a stage exists is therefore an empirical question.

Homonyms provide a controlled assay because the same lexical form can support sharply different interpretations. If hidden states primarily preserve word form, occurrences of *bank* should remain similar across financial and riverside contexts. If contextual sense is expressed, the two groups should become geometrically distinguishable. The target form is held constant, but the surrounding sentences still differ in vocabulary, syntax, topic, length, and target position. The dependent measure is therefore *sense-conditioned contextual separability*, not an isolated or exhaustive semantic feature.

A. Locating usable sense information

A claim that semantics occupies a particular layer depends on how “semantic information” is operationalised. This study compares two geometric criteria. A held-out nearest-centroid margin measures whether an individual sentence lies closer to the centroid of its authored sense than to the opposing sense. The Generalized Discrimination Value (GDV) instead

summarises labelled within-sense and between-sense distances across the complete representation space [7]. The former emphasises reliable item-level decisions; the latter emphasises global class geometry without fitting a classifier.

H1 tests whether selecting a layer from profiling data improves held-out sense decodability relative to using the final layer. H2 asks whether such a choice transfers to an unseen homonym and whether margin- and GDV-based criteria identify the same depth region. Nested and cross-word held-out evaluation are used because an apparent semantic layer is informative only if it generalises beyond the observations or word that selected it [8].

B. Layer depth, token position, and available context

Layer depth is not equivalent to reading time. Three axes must be distinguished:

- 1) Layer depth indexes successive transformations within one forward pass.
- 2) Sequence position determines which context is causally available at a particular token.
- 3) Incremental input is studied by rerunning the model on progressively longer prefixes.

This distinction is especially important when comparing bidirectional encoders with causal decoders. At a homonym position, an encoder can use both preceding and following context, whereas a causal decoder can never incorporate a later resolver there, regardless of depth. A later decoder token can nevertheless attend to both the homonym and the resolver. Consequently, failure to decode a sense at one token does not establish that the completed sentence lacks that information elsewhere.

H3 manipulates whether disambiguating context appears before or after the homonym and therefore whether it is causally available at that position. H4 then asks whether sense information unavailable at the decoder homonym becomes locally decodable at a later sentence position after the resolver has been encountered. Together, these analyses prevent a context-availability effect from being misinterpreted as a semantic-layer effect.

C. Baseline lean and incremental context conflict

Even without explicitly disambiguating context, a homonym may lie closer to one of the two profiled sense regions. H0 measures this baseline asymmetry for bare words and ambiguous carrier sentences. These states provide a starting point for interpreting later updating: identical contextual evidence may lead to different endpoints when representations begin from different positions along the sense axis.

H5 tests whether that representation changes when new context contradicts an earlier interpretation. Human comprehension is incremental, and later disambiguating information can produce processing costs and leave traces of an initial garden-path interpretation [9, 10]. Violation-sensitive neural responses such as the N400 likewise show that incoming language is evaluated relative to semantic expectations [2, 11]. These findings motivate a model-internal context-conflict test,

but the present experiment does not measure human reading time, neural activity, or conscious interpretation.

Models are therefore rerun on progressively longer prefixes that first prime one sense and later resolve the homonym toward the other. A fixed sentinel provides a common readout across stages. The analysis distinguishes movement toward the resolved-sense region, categorical crossing of the centroid boundary, and the remaining difference from matched non-conflicting controls. It tests geometric updating and persistence without claiming equivalence to human garden-path reanalysis.

D. Research questions and analysis structure

The study addresses three questions.

RQ1 Where is contextual sense information most accessible?

Does sense information become especially clear at a reproducible depth, or does its apparent location vary across models, homonyms, and geometric criteria?

H0 measures the baseline sense lean of ambiguous inputs. H1 tests whether an intermediate layer provides a more reliable sense readout than the final layer. H2 tests whether layer selection transfers across homonyms and geometric criteria.

RQ2 How does sense accessibility depend on readout position and available context?

Does sense decodability depend only on layer depth, or also on the context available at the selected token position?

H3 compares disambiguating information placed before and after the homonym. H4 tests whether sense information becomes accessible at a later sentence position after the resolver has been processed.

RQ3 How flexibly is an established interpretation revised?

When later context supports a different sense, does the representation move toward that sense, cross into its geometric region, and approach the endpoint of a non-conflicting context?

H5 measures incremental updating and the remaining effect of preceding conflicting context.

Together, these analyses test whether contextual meaning has a reproducible location in transformer depth and how its representation changes with context. Agreement across models, homonyms, criteria, and token positions would support a stable semantic stage. Variation across these dimensions would support a distributed and path-dependent account.

II. DATA AND METHODS

The study analysed three dimensions of transformer representations. H1–H2 varied layer depth within one forward pass. H3–H4 varied token position and the context available at that position. H5 varied the supplied context across forward passes by rerunning each model on longer prefixes.

TABLE I
THE SEVEN HOMONYMS AND THEIR OPERATIONAL SENSE CONTRASTS.

Word	Sense 0	Sense 1
bank	river / embankment	financial institution
bark	dog vocalisation	tree covering
bat	sports equipment	flying mammal
crane	large wading bird	lifting machine
match	sports contest	fire-starting stick
spring	season	water source
pitch	sports field	business proposal

TABLE II
SAMPLE PROFILING STIMULI FOR *bank*. THE FULL SET CONTAINS 20 SENTENCES PER SENSE. IN EACH H1 OUTER FOLD, ONE SENTENCE IS HELD OUT. THE REMAINING 39 SENTENCES DETERMINE THE SELECTED LAYER AND THE SENSE CENTROIDS.

Sense	Operational label	Example profiling sentence
0	Riverside	There’s a steep bank at the bend of the river.
0	Riverside	Fishermen sat quietly on the bank waiting for a bite.
1	Financial	She went to the bank to open a new account.
1	Financial	The bank manager approved her loan application.

A. Shared design

a) *Operational sense contrasts.*: Seven English homonyms defined seven binary sense contrasts (Table I). The labels cover the tested senses only. *Spring* and *pitch*, for example, have additional readings.

An eighth candidate, *light*, was excluded because its intended contrast paired a noun sense (*illumination*) with an adjective sense (*low weight*). This would couple sense with grammatical category in the ambiguity and resolution items.

All stimuli were authored for this study. The homonym remained constant within each contrast. The surrounding contexts varied in vocabulary, topic, syntax, length, and sometimes target position. The measures therefore capture *sense-conditioned contextual separability* and *local sense decodability* within the seven operational contrasts.

b) *Profiling sentences.*: The profiling set contained 20 unambiguous sentences per sense and homonym, giving 280 sentences in total (Table II). Target occurrences were identified with word-boundary-aware matching. The target could appear in its singular form or as a regular plural.

When tokenisation split a target into several subwords, their hidden states were averaged. For example, Qwen2.5 split *Cranes* in “Cranes migrate thousands of miles ...” into *C*, *ran*, and *es*. Their mean represented the target occurrence. These homonym-position states defined the reference centroids used throughout the study.

For H4, the same profiling pass also retained the final non-special, non-padding state. This token was the sentence-final period in every profiling and H4 sentence. The period-position states defined a common endpoint geometry for the final-position readout.

c) *Models.*: The sample included four bidirectional encoders: ModernBERT-large, DeBERTa-v3-large, RoBERTa-

large, and XLM-RoBERTa-large. It also included four causal decoders: Qwen2.5-3B, Qwen2.5-7B, Mistral-Nemo-Base-2407, and OLMo-2-1124-7B. All models were open-weight base checkpoints, which allowed hidden-state extraction at every layer without instruction tuning.

The groups also differed in parameter count, training objective, training data, tokenizer, and depth. Architecture summaries are therefore descriptive. The paired context-order manipulation in H3 directly tests the effect of causal context availability.

B. Representational measures

a) *Adequacy margin.*: For a hidden state h_l at layer l , let $c_{0,l}$ and $c_{1,l}$ denote the profiling centroids for the two senses at the relevant readout position. For an observation with authored sense label $s \in \{0, 1\}$, the Euclidean adequacy margin is

$$M_l(h, s) = \|h_l - c_{1-s,l}\|_2 - \|h_l - c_{s,l}\|_2. \quad (1)$$

A positive value places the observation closer to its authored-sense centroid. A negative value places it closer to the opposing centroid. Zero marks the nearest-centroid boundary. The sign is assigned relative to each observation’s authored sense, so positive values always indicate correct-sense proximity.

The raw margin was normalised by the distance between the two centroids:

$$\widehat{M}_l(h, s) = \frac{M_l(h, s)}{\|c_{1,l} - c_{0,l}\|_2}. \quad (2)$$

The reverse triangle inequality gives $-1 \leq \widehat{M}_l \leq 1$. The two centroids attain the endpoint values, and zero remains the decision boundary. Normalisation reduces the effect of activation scale across models and layers. Binary adequacy uses the sign $M_l > 0$. Magnitude comparisons use $\widehat{M}_l$.

b) *Centroid estimation and layer use.*: H1 and H2 scored each profiling sentence against centroids estimated from the remaining sentences. H1 also selected the layer inside each outer leave-one-sentence-out fold. Layer selection and centroid estimation used only the outer training set.

H0 and H3–H5 used centroids and layers estimated from the separate profiling set. Their reference geometry came only from profiling sentences.

c) *Uncertainty and thresholds.*: Architecture summaries used model–word aggregates as their evidential units. Contrasts in H1–H3 received 95% intervals from 20,000 crossed bootstrap resamples of models and homonyms. H4 reports pooled counts and proportions for its conditional transitions. Thresholds such as $|\widehat{M}_l| > 0.3$ appear only in descriptive sensitivity summaries. Continuous margins and the zero decision boundary remain the primary outcomes.

C. H0: baseline sense lean

H0 compared a bare homonym with five ambiguous carrier sentences (Table III). Each input completed a full forward pass. The homonym state was then read at the independently

TABLE III
H0 INPUTS FOR *bank*. THE BARE WORD AND FIVE CARRIERS PRESERVE AMBIGUITY BETWEEN THE RIVERSIDE AND FINANCIAL SENSES.

Condition	Model input
Bare word	bank
Neutral carrier 1	The bank was unstable.
Neutral carrier 2	They examined the bank.
Neutral carrier 3	The bank had changed.
Neutral carrier 4	The bank needed protection.
Neutral carrier 5	The bank became difficult to access.

selected layer for that model–word pair. This design separates the supplied context from the depth of the readout.

The two profiling centroids at the same selected layer defined the sense axis. H1 separately used layer 0 as a diagnostic for surface-form and position differences.

The signed lean was

$$B_l(h) = \frac{\|h_l - c_{1,l}\|_2 - \|h_l - c_{0,l}\|_2}{\|c_{1,l} - c_{0,l}\|_2}. \quad (3)$$

Positive values lean toward sense 0. Negative values lean toward sense 1. The two label-specific margins for one carrier are exact opposites, so each carrier contributed one signed state. H0 reports the bare-word lean, the mean carrier lean, its absolute magnitude, the bare-to-carrier shift, and carrier-direction consistency. The count with $|B_l| > 0.3$ provides a sensitivity summary.

D. H1: layer depth and held-out adequacy

H1 tested whether a profiling-selected contextual layer improved held-out sense adequacy relative to the final layer. Each profiling sentence was scored at every contextual layer with leave-one-sentence-out centroids. Selection started at layer 1.

Layer 0 was retained as a surface diagnostic. It precedes self-attention, so its label separation reflects properties such as token form, tokenisation, and target position.

Performance used nested leave-one-sentence-out evaluation. Each outer fold held out one profiling sentence. The remaining sentences selected the layer with the highest inner leave-one-out adequacy. Mean normalised margin broke ties. The held-out sentence was then scored at the selected and final layers with centroids estimated from the outer training set.

A separate full-profile layer was estimated for H0 and H3–H5, which used independent stimuli. The main H1 contrast was selected-layer minus final-layer adequacy across model–word cells. Modal layer and relative depth described the variation in layer choice.

E. H2: cross-word layer-selection strategies

H2 tested whether a layer choice transferred to a new homonym. Each homonym was held out in turn. The remaining six homonyms selected a layer, and the held-out homonym was evaluated with its own leave-one-sentence-out centroids.

Four strategies were compared:

TABLE IV
ONE H3 CARRIER SET FOR *bank*. EACH CARRIER APPEARS WITH ONE CLAUSE PER SENSE AND IN BOTH CLAUSE ORDERS.

Condition	Intended sense	Sentence
L	Riverside	After weeks of heavy rain, the bank was unstable .
R	Riverside	The bank was unstable after weeks of heavy rain.
L	Financial	After weeks of market panic, the bank was unstable .
R	Financial	The bank was unstable after weeks of market panic.

- 1) *GDV*: the layer with the most negative mean GDV across the six profiling homonyms;
- 2) *adequacy-based cross-word*: the layer with the highest mean leave-one-out adequacy across those homonyms;
- 3) *last*: the final layer; and
- 4) *oracle*: the best layer for the held-out homonym, used as a descriptive ceiling.

For a layer with D -dimensional states, each feature was standardised as $x'_{nd} = \frac{1}{2}(x_{nd} - \mu_d)/\sigma_d$. GDV was then computed as

$$\text{GDV} = \frac{\bar{d}_{\text{intra}} - \bar{d}_{\text{inter}}}{\sqrt{D}}, \quad (4)$$

where $\bar{d}_{\text{intra}}$ is the mean within-sense distance and $\bar{d}_{\text{inter}}$ is the mean between-sense distance. More negative values indicate stronger average class separation.

The comparison reports held-out adequacy, adequacy regret relative to the oracle, absolute layer distance from the oracle, and the proportion of selected layers within two positions of the oracle. Crossed model–word bootstrap intervals compare each transferable strategy with the final layer.

F. H3: causal availability at the homonym position

H3 manipulated whether the decisive clause appeared before or after the homonym. Each homonym had five ambiguous carriers. Every carrier appeared with one clause for each sense and in both clause orders (Table IV). This produced 20 sentences per homonym and 140 sentences in total.

The homonym state was read at the profiling-selected layer and scored against homonym-position centroids. Each L sentence was paired with its R reordering. The primary outcome was the paired L-minus-R change in normalised margin. Effects were first averaged within model–word cells. The architecture contrast was the decoder-minus-encoder difference in this paired effect.

For causal decoders, the two R sentences for one carrier share the same prefix at the homonym and have opposing labels. Their mean adequacy is therefore 0.500 by construction. The main estimates are the L-minus-R margin and its architecture interaction.

G. H4: sequence-position decodability

H4 used the ten R-condition sentences per homonym. It compared the homonym and the sentence-final period within the same completed forward pass. The analysis contained 560 observations. Both positions were read at the same selected layer, so layer depth stayed fixed.

The homonym state was scored against homonym-position profiling centroids. The period state was scored against period-position profiling centroids. A cross-position diagnostic also scored the period state against the homonym centroids.

The gain rate was

$$P(\text{final adequate} \mid \text{target inadequate}),$$

and the loss rate was

$$P(\text{final inadequate} \mid \text{target adequate}).$$

H4 reports pooled counts and proportions for both transitions. The comparison combines the additional context available at the endpoint with the change in readout position. Token identity, clause length, and cue-to-endpoint distance also vary across the two readouts.

H. H5: incremental context-conflict updating

H5 reran each model on progressively longer prefixes while keeping the readout token fixed.

The final set contained 98 items, with 14 items per homonym. Seven items per homonym primed sense 0 and resolved to sense 1. The other seven used the opposite direction. Each item had a matched non-conflicting control with the same homonym and resolver. All items followed the order prime, homonym, resolver.

The sentinel *probe* was appended in a new paragraph after three prefixes: the prime, the homonym, and the resolver. The model was rerun for each prefix. The sentinel state was read at the same profiling-selected layer. Independent profiling sentences with the same sentinel defined its sense centroids.

The three stages therefore shared the sentinel token, terminal readout role, selected layer, and sense axis. Prefix growth changed the sentinel’s absolute sequence position. The stage

comparison captures the combined effect of added context and this position shift.

The primary outcome was

$$\Delta \widehat{M} = \widehat{M}_{\text{resolution}} - \widehat{M}_{\text{homonym}}.$$

A positive value indicates movement toward the authored resolved sense. The conditional transition rate was

$$P(\text{resolution correct} \mid \text{homonym primed}).$$

The matched control measured the endpoint cost of preceding conflict. The resolver-only input measured the endpoint associated with the resolver’s lexical content. Thresholds of 0.1 and 0.3 provided sensitivity summaries. The continuous paired change remained the primary outcome.

I. Exploratory analyses

Two post-hoc analyses examined relationships across the primary analyses.

First, H0 baseline lean was compared with H1 adequacy and H5 updating. The H0–H1 analysis used all 56 model–word cells. H5 was also separated by conflict direction, giving 112 model–word–direction cells. Transition rates were defined for 107 cells. Five cells had an empty primed denominator. Spearman correlations used these aggregated cells. The analyses remained descriptive and report raw correlations.

Second, H5 outcomes were compared between encoders and decoders. Outcomes were first averaged within each of the 56 model–word cells. Decoder-minus-encoder differences and 95% intervals used 20,000 crossed bootstrap resamples of models and homonyms. These comparisons describe the sampled architecture groups.

TABLE V
ONE H5 *bank* ITEM. THE PRIME SUPPORTS THE FINANCIAL SENSE AND THE RESOLVER SELECTS THE RIVERSIDE SENSE. EACH ROW IS A SEPARATE MODEL INPUT. THE INDENTED *probe* IS THE FINAL PARAGRAPH OF THAT INPUT.

Input condition	Function	Exact input supplied to the model
Prime stage	Financial prime only	She finished her loan application and walked to the <i>probe</i>
Homonym stage	Prime plus ambiguous word	She finished her loan application and walked to the bank <i>probe</i>
Resolution stage	Conflict through the resolver	She finished her loan application and walked to the bank to sit by the river <i>probe</i>
Matched control	Non-conflicting context through the same resolver	After following the riverside path, she walked to the bank to sit by the river <i>probe</i>
Resolver in isolation	Lexical resolver baseline	river <i>probe</i>

III. RESULTS

A. H0: baseline sense lean

Across 56 model–word cells, the mean absolute carrier lean was 0.339. Mean direction consistency across the five carriers was 0.882. The descriptive threshold $|B_l| > 0.3$ was exceeded by 159 of 280 carrier states. Encoders contributed 65 of 140 cases and decoders contributed 94 of 140 (Table VI).

TABLE VI

H0 CARRIER LEAN BY ARCHITECTURE. MEAN $|B_l|$ GIVES THE ABSOLUTE LEAN TOWARD EITHER CENTROID. DIRECTION CONSISTENCY GIVES THE LARGER SIGN PROPORTION AMONG NON-ZERO CARRIER LEANS. THE FINAL COLUMN COUNTS STATES WITH $|B_l| > 0.3$.

Architecture	Mean $ B_l $	Direction consistency	$ B_l > 0.3$
Encoder	0.279	0.864	65/140
Decoder	0.398	0.900	94/140
All models	0.339	0.882	159/280

The largest model-specific leans were Qwen2.5-3B toward tree covering for *bark* ($B_l = -0.657$), Qwen2.5-3B toward water source for *spring* ($B_l = -0.617$), and OLMo-2-1124-7B toward water source for *spring* ($B_l = -0.594$). All three cells had complete carrier-direction agreement.

Several directions also recurred across models. Seven of eight models leaned toward the financial sense of *bank*. All eight leaned toward tree covering for *bark*. Seven leaned toward water source for *spring*, and seven toward the sporting sense of *match*. *Bat* had an across-model mean near zero and the lowest direction consistency (Figure 1).

Bare-word and carrier directions agreed in 28 of 56 cells. Agreement was highest for *bank* (7/8 models) and *match* (5/8), and lowest for *spring* (2/8). Carrier context often moved the mean lean farther from zero. It could also weaken or reverse the bare-word direction. For Qwen2.5-7B, bare *crane* strongly favoured bird, while the carriers consistently favoured machine with a smaller mean magnitude. The contextual baseline therefore depended on both the homonym and its underspecified carrier.

B. H1: nested layer selection

Layer-0 adequacy was 0.500 across the 56 model–word cells. Word means ranged from 0.238 for *bank* to 0.675 for *crane*. Because layer 0 precedes self-attention, these differences reflect surface properties of the profiling sets. Target position differed between senses for several words. For example, *bat* followed an average of 4.1 words in sense 0 and 2.0 words in sense 1. The corresponding values for *pitch* were 5.2 and 2.45.

Across 2,240 outer folds, nested selection reached 0.917 adequacy. The final layer reached 0.893, giving a pooled difference of 2.37 percentage points (Table VIII). The difference was 0.63 points for encoders and 4.11 points for decoders. The crossed model–word effect was 0.0237 with a 95% interval of $[-0.0058, 0.0598]$.

Full-profile layer choices differed by architecture. Encoder choices had a mean relative depth of 0.762 and a median of 0.866. All 28 encoder–word cells selected at or above one

third of model depth. One cell, DeBERTa-v3-large on *spring*, selected the one-third boundary. Decoder choices had a mean relative depth of 0.494 and a median of 0.366. Thirteen of 28 decoder cells selected the first third, and six selected the final layer. Encoder means were later for every homonym, although the model ranges overlapped.

Model effects also varied (Figure 2). Selection improved Qwen2.5-3B by 9.29 points and Mistral-Nemo-Base-2407 by 6.79 points. It reduced adequacy slightly for Qwen2.5-7B, ModernBERT-large, and RoBERTa-large. The final layer was therefore a strong default for some models and a weaker choice for others.

C. H2: cross-word selection and the GDV objective

GDV-selected layers reached 0.892 held-out adequacy, compared with 0.893 at the final layer (effect = -0.0009 , 95% CI $[-0.0366, 0.0308]$). Adequacy-based cross-word selection reached 0.914 (effect = 0.0210 , 95% CI $[-0.0013, 0.0496]$). The held-out-word oracle reached 0.930 (effect relative to last = 0.0371 , 95% CI $[0.0112, 0.0728]$) (Table IX).

Figure 3 gives a separate full-profile comparison within each model–word cell. It is distinct from the leave-one-word-out transfer analysis.

The two cross-word selectors chose the same layer in 7 of 56 folds. They chose layers within two positions in 14 folds, and their median separation was four layers.

The adequacy difference was concentrated among decoders. Encoder adequacy was 0.960 under GDV, 0.966 under adequacy-based selection, and 0.960 at the final layer. Their oracle reached 0.980. Decoder adequacy was 0.824 under GDV, 0.862 under adequacy-based selection, and 0.826 at the final layer. Their oracle reached 0.879.

Transfer also varied by homonym. *Crane* had the lowest adequacy under GDV-based selection (0.838). *Spring* and *bark* had the largest oracle regret under GDV, at 0.075 and 0.072. *Bank* was stable under both criteria (0.900 and 0.903). GDV exceeded adequacy-based selection for *match* (0.900 versus 0.881) and slightly for *pitch* (0.888 versus 0.884). For *bat*, the corresponding means were 0.925 and 0.959.

GDV and adequacy summarise different features of a layer. GDV averages standardised within-class and between-class distances across all observations. Adequacy tests whether each held-out sentence lies closer to the correct centroid in the original hidden space. A layer can improve average pairwise separation while leaving several sentences on the wrong side of the centroid boundary. Those boundary errors directly reduce adequacy and may have little effect on GDV. GDV therefore provides a less direct estimate of item-level decodability.

The Qwen2.5-7B *bat* case shows this difference (Figures 4 and 5). Layer 26 had the more negative GDV (-0.0634 versus -0.0505), while layer 3 had higher leave-one-out adequacy (0.90 versus 0.60). The PCA projection shows the change in class geometry. The full-space margins show which sentences crossed the decision boundary.

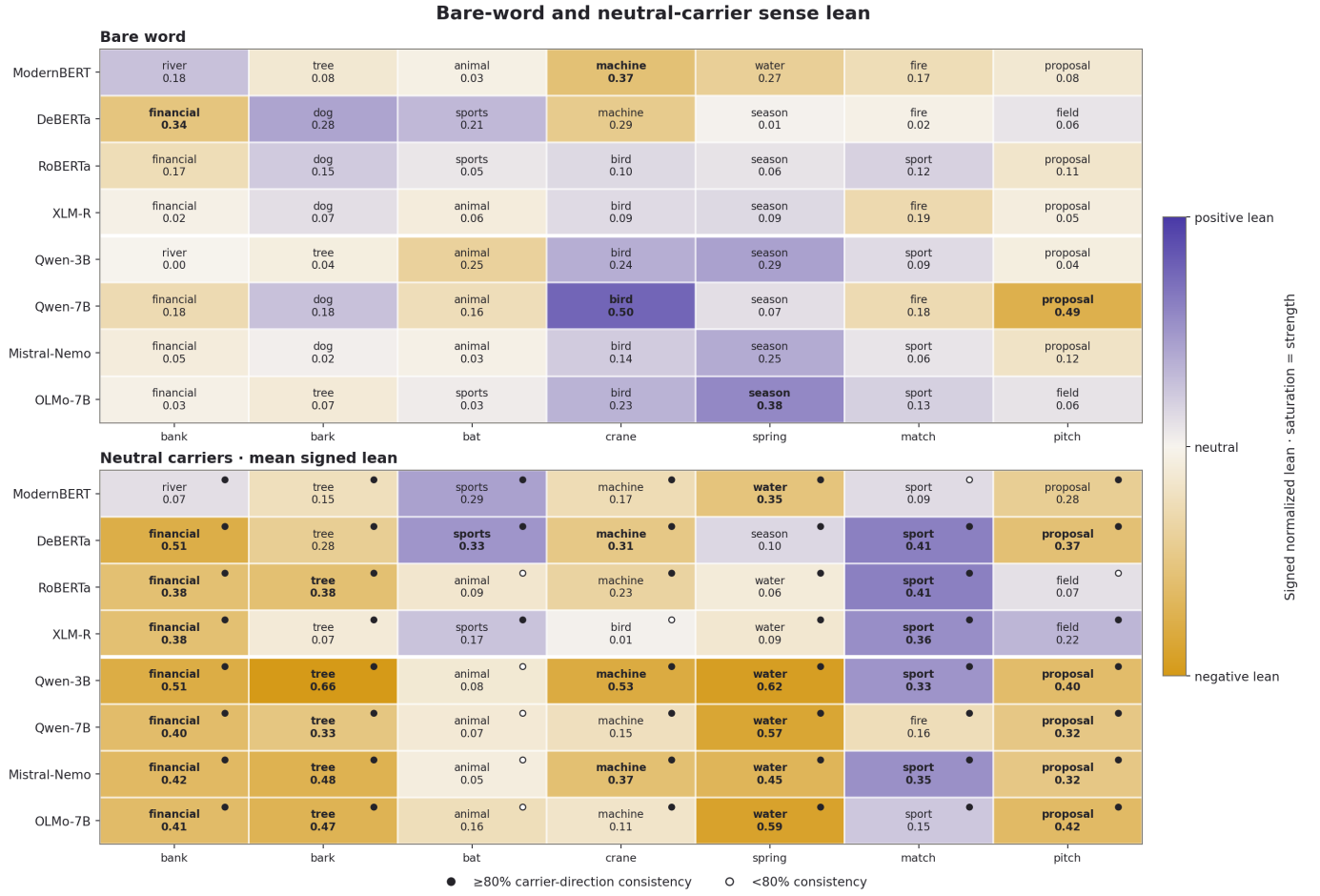


Fig. 1. Bare-word and carrier sense lean by model and homonym. Each cell names the closer sense. Colour saturation shows $|B_l|$. In the carrier panel, a filled dot marks agreement in at least four of five carriers.

TABLE VII
CARRIER AUDIT FOR THE FOUR HOMONYMS WITH THE LARGEST ABSOLUTE ACROSS-MODEL MEAN LEAN. MEAN B_l AVERAGES EIGHT MODELS AND FIVE CARRIERS.

Word	Mean direction	Mean B_l	Neutral carriers
<i>bank</i>	Financial	−0.367	The bank was unstable. They examined the bank. The bank had changed. The bank needed protection.
<i>bark</i>	Tree	−0.350	The bank became difficult to access. The bark drew attention. They studied the bark. The bark was unusual. The bark changed noticeably.
<i>spring</i>	Water	−0.329	The bark became the focus. The spring was unusually dry. They monitored the spring. The spring changed quickly. The spring became a concern.
<i>match</i>	Sport	+0.242	The spring was hard to assess. The match was difficult to start. The match ended quickly. They checked the match. The match caused trouble. The match was watched closely.

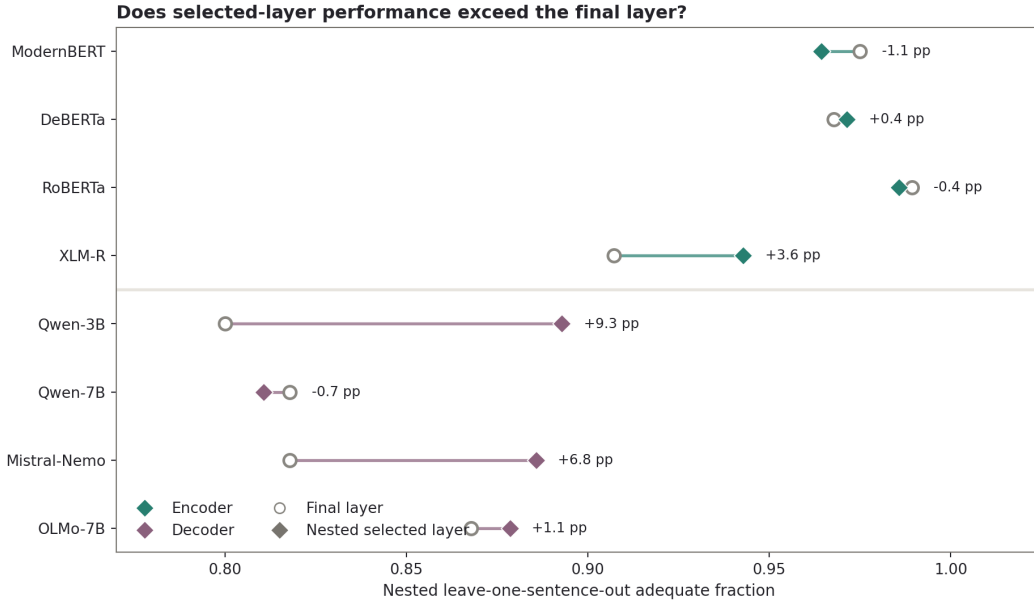


Fig. 2. Model-level H1 adequacy at the nested selected layer and final layer. Labels give the selected-minus-final difference in percentage points.

Do GDV and M_I select the same layer?							
Encoders							
ModernBERT · 2/8	24→25	L25	14→25	26→24	L24	28→25	27→25
DeBERTa · 3/8	9→8	L10	L9	L10	8→11	18→10	23→12
RoBERTa · 6/8	L21	L19	L19	L15	L22	23→19	L21
XLM-R · 3/8	L23	21→11	L21	21→23	13→22	20→23	L22
Decoders							
Qwen-3B · 3/8	L7	L14	32→14	L32	32→12	15→12	15→14
Qwen-7B · 2/8	28→3	28→11	L28	L28	3→2	28→10	28→3
Mistral-Nemo · 4/8	L4	L7	L7	7→1	6→1	L7	21→24
OLMo-7B · 3/8	L6	8→7	L8	7→1	7→1	L22	11→22
	bank 5/8 agree	bark 5/8 agree	bat 6/8 agree	crane 4/8 agree	spring 2/8 agree	match 2/8 agree	pitch 2/8 agree
	same selected layer different layers ($M_I \rightarrow$ GDV)						

Fig. 3. Full-profile layer choices under nearest-centroid adequacy and GDV. Each cell shows the shared layer or the two selected layers. The criteria agree in 26 of 56 model-word cells.

D. H3: causal availability at the homonym

Encoder margins changed little across clause order. Their mean margin was 0.360 in L and 0.358 in R, giving an L-minus-R effect of 0.0026. Decoder margins changed from 0.342 in L to approximately zero in R, giving an effect of 0.3420. The decoder-minus-encoder interaction was 0.3394 with a 95% interval of [0.2358, 0.4355] (Table X).

The decoder R adequacy of 0.500 confirmed the construction

check. All four decoders showed a large clause-order effect, while each encoder remained near zero (Figure 6).

The architecture interaction was positive for every homonym. It was largest for *crane* (0.492), *match* (0.376), and *bat* (0.357), and smallest for *spring* (0.161). Encoder L-minus-R changes stayed near zero for most words. *Match* had the largest encoder change at 0.076.

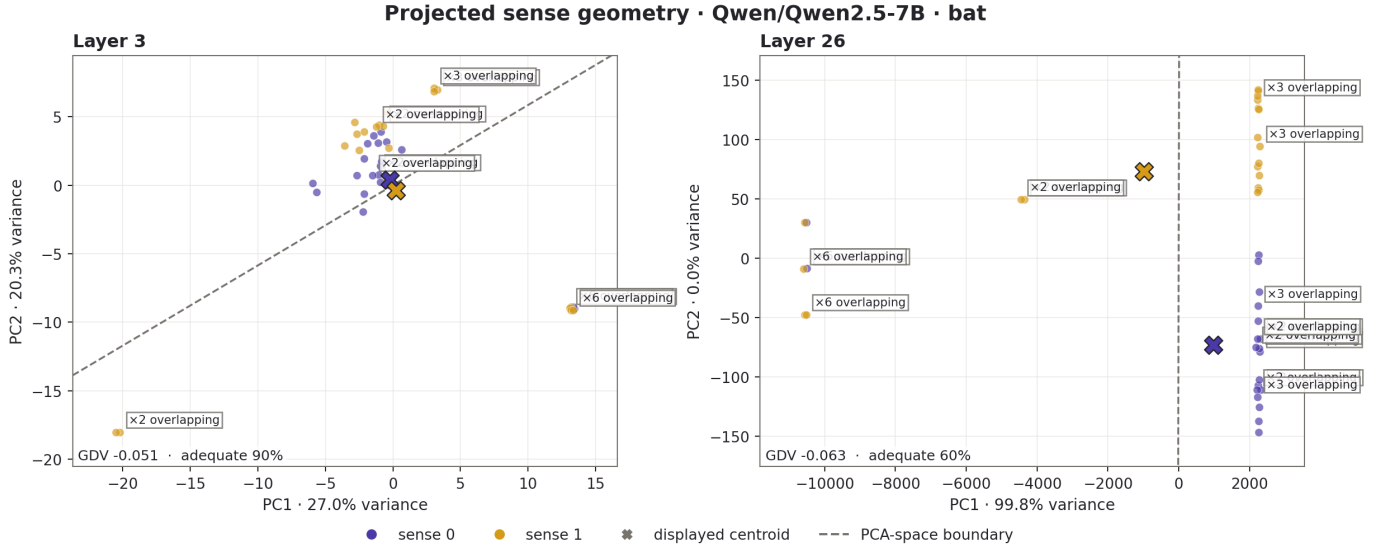


Fig. 4. PCA projections of Qwen2.5-7B *bat* states at layers 3 and 26. The dashed boundary uses the two plotted dimensions. Reported GDV and adequacy use the full hidden space.

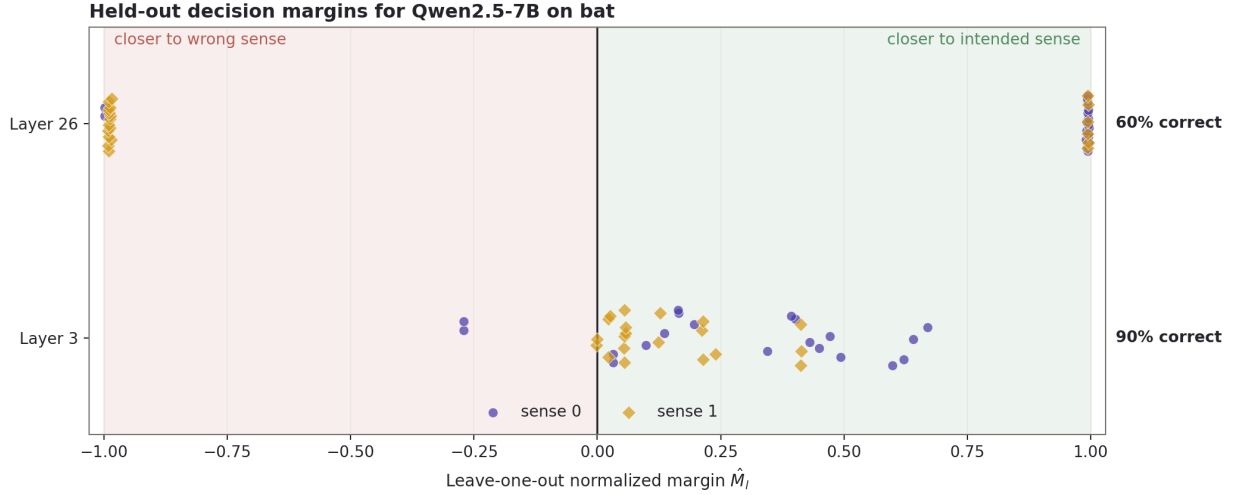


Fig. 5. Full-space leave-one-out margins for Qwen2.5-7B on *bat*. Layer 3 assigns 90% of sentences correctly. Layer 26 assigns 60% correctly and has the more favourable GDV.

TABLE VIII
HELD-OUT H1 ADEQUACY UNDER NESTED LAYER SELECTION.
DIFFERENCE IS SELECTED MINUS LAST.

Architecture	Outer folds	Selected	Last	Difference
Encoder	1,120	0.966	0.960	0.006
Decoder	1,120	0.867	0.826	0.041
All models	2,240	0.917	0.893	0.024

TABLE IX
H2 CROSS-WORD LAYER-SELECTION PERFORMANCE ACROSS 56 HELD-OUT
MODEL-WORD FOLDS. REGRET AND LAYER DISTANCE ARE MEASURED
FROM THE HELD-OUT ORACLE. WITHIN 2 IS THE PROPORTION SELECTED
WITHIN TWO LAYERS OF THE ORACLE.

Strategy	Adequacy	Oracle regret	Layer distance	Within 2 layers
GDV	0.892	0.038	4.62	0.643
Adequacy-based				
cross-word	0.914	0.016	6.62	0.339
Last layer	0.893	0.037	11.96	0.250
Oracle	0.930	0.000	0.00	1.000

E. H4: local decodability across sequence positions

Decoder adequacy was 0.500 at the homonym and 0.761 at the sentence-final period. Among decoder observations that were inadequate at the homonym, 67.9% became adequate at

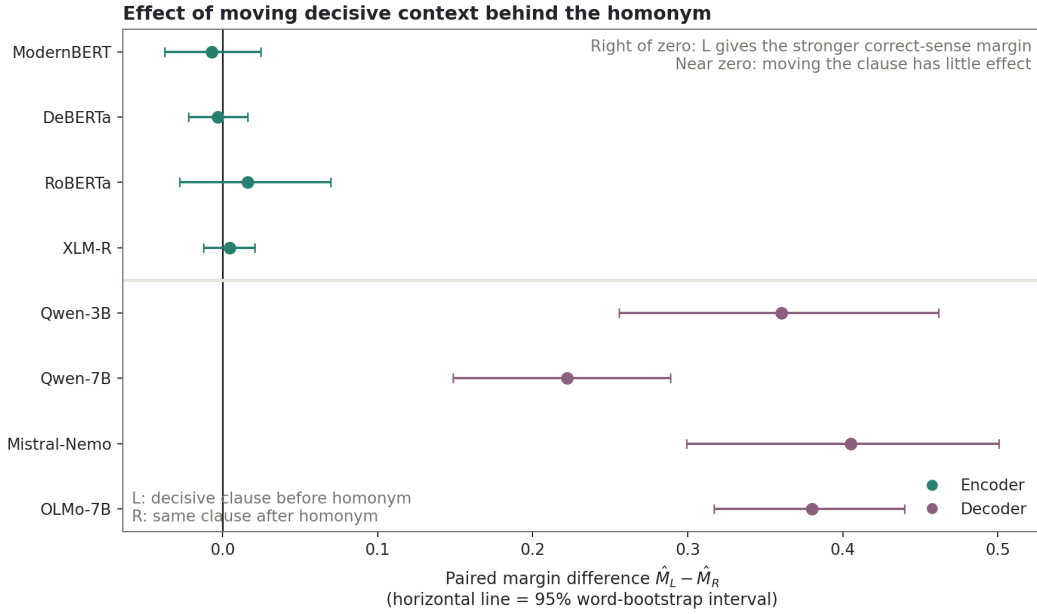


Fig. 6. Per-model H3 change from L to R. Points show mean $\hat{M}_L - \hat{M}_R$ across homonyms. Intervals are 95% word-bootstrap intervals.

TABLE X

H3 MARGINS AND ADEQUACY WHEN THE DECISIVE CLAUSE APPEARS BEFORE (L) OR AFTER (R) THE HOMONYM.

Architecture	$\hat{M}_L$	$\hat{M}_R$	Adequacy L	Adequacy R
Encoder	0.360	0.358	0.843	0.861
Decoder	0.342	-0.0002	0.918	0.500

TABLE XI

H4 CHANGE IN LOCAL SENSE DECODABILITY FROM THE HOMONYM TO THE SENTENCE-FINAL PERIOD. GAIN AND LOSS USE THE CONDITIONAL DENOMINATORS SHOWN.

Architecture	Target	Final	Gain	Loss
Encoder	0.861	0.668	16/39 = 0.410	70/241 = 0.290
Decoder	0.500	0.761	95/140 = 0.679	22/140 = 0.157

the endpoint. Among adequate homonym observations, 15.7% became inadequate at the endpoint. Encoder adequacy changed from 0.861 to 0.668, with a gain rate of 41.0% and a loss rate of 29.0% (Table XI).

Decoder endpoint adequacy exceeded 0.500 for every homonym. Conditional gain was highest for *crane* (0.950), followed by *pitch* (0.800), *bank* (0.750), and *bark* (0.750). *Spring* had a gain rate of 0.200 and endpoint adequacy of 0.550. Encoder endpoint adequacy was lower than homonym adequacy for every word.

The cross-position diagnostic scored period states against homonym centroids. Its overall adequacy was 0.516. The homonym and period therefore used different local centroid geometries.

F. H5: fixed-sentinel incremental updating

The balanced design produced 784 model-item observations. The sentinel mean was -0.080 after the prime, -0.129 after the homonym, and -0.001 after the resolver. The resolver-minus-homonym change was $+0.128$. In total, 605/784 observations (77.2%) moved toward the resolved sense. Among the 580 observations on the primed side at the homonym, 234 (40.3%) crossed the boundary by the resolver. Table XII and Figure 7 show the architecture trajectories.

Both architecture groups moved toward the resolved sense. Decoders had the larger mean change (0.137 versus 0.119), the lower resolver endpoint (-0.022 versus 0.021), and the lower transition rate (38.7% versus 42.4%). Their stronger homonym-stage priming placed more observations in the conditional denominator and farther from the boundary.

The post-hoc decoder-minus-encoder difference in update magnitude was 0.018 with a 95% interval of $[-0.048, 0.080]$. The decoder resolver endpoint was lower by 0.043 ($[-0.087, -0.001]$). Its conflict-minus-control difference was lower by 0.084 ($[-0.165, -0.010]$). These estimates describe the sampled architecture groups.

Decoder updates were larger for five homonyms. Encoders moved farther for *bark* and *bat*. Encoder resolver endpoints were higher for six homonyms, while decoders were higher for *pitch*. The mean update was positive for every model, ranging from 0.074 for ModernBERT to 0.186 for DeBERTa. Fifty-four of 56 model-word cells had a positive mean update. The exceptions were XLM-R on *crane* (-0.009) and *spring* (-0.047).

Every homonym had a positive mean update. Values ranged from 0.068 for *spring* to 0.181 for *bank*. Transition rates ranged from about 31% for *crane* and *spring* to 56.6% for

TABLE XII

H5 MARGINS AND UPDATING BY ARCHITECTURE. TRANSITION IS THE CROSSING RATE AMONG OBSERVATIONS ON THE PRIMED SIDE AT THE HOMONYM.

Arch.	Prime	Homonym	Resolver	Δ	Positive Δ	Transition
Encoder	-0.062	-0.098	0.021	0.119	284/392 = 72.4%	111/262 = 42.4%
Decoder	-0.098	-0.159	-0.022	0.137	321/392 = 81.9%	123/318 = 38.7%

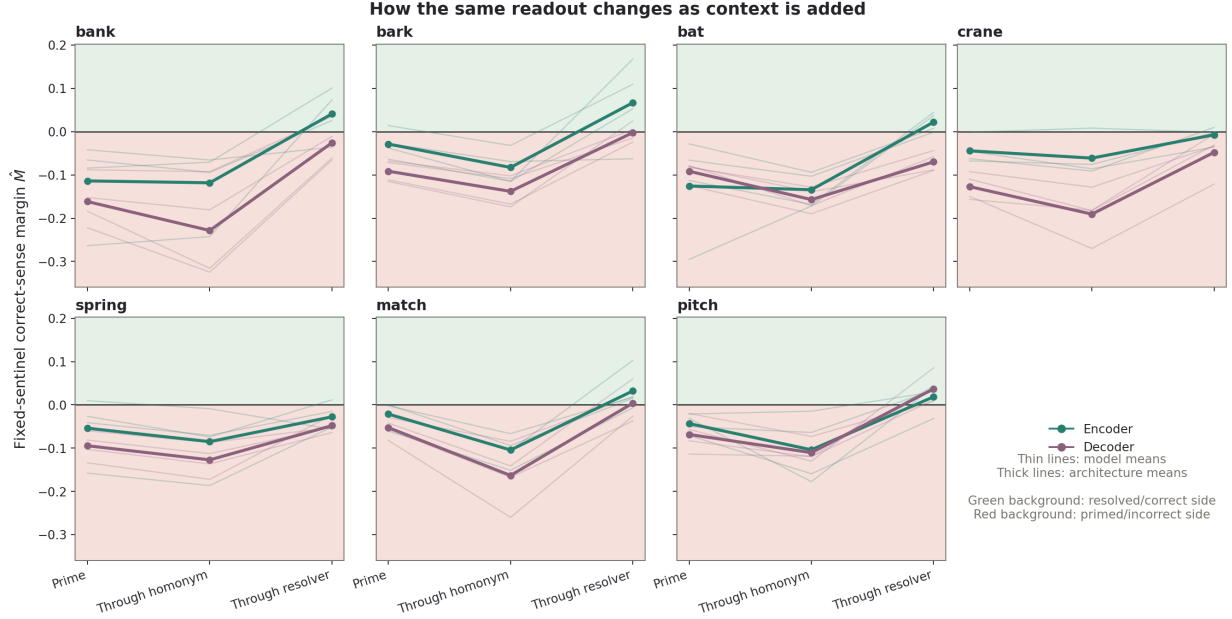


Fig. 7. Fixed-sentinel margin across the prime, homonym, and resolver stages. Thin lines show model means. Thick lines show encoder and decoder means.

TABLE XIII

WORD-LEVEL H5 OUTCOMES. MEAN Δ IS THE RESOLVER-MINUS-HOMONYM CHANGE. TRANSITION IS THE CROSSING RATE AMONG INITIALLY PRIMED OBSERVATIONS.

Word	Mean Δ	Positive Δ	Transition	Conflict-control
bank	0.181	93/112 (83.0%)	38/90 (42.2%)	-0.162
bark	0.143	101/112 (90.2%)	35/83 (42.2%)	-0.116
bat	0.122	81/112 (72.3%)	31/78 (39.7%)	-0.135
crane	0.098	80/112 (71.4%)	27/87 (31.0%)	-0.174
spring	0.068	79/112 (70.5%)	24/77 (31.2%)	-0.099
match	0.152	97/112 (86.6%)	32/82 (39.0%)	-0.158
pitch	0.135	74/112 (66.1%)	47/83 (56.6%)	-0.135

pitch. Direction also mattered. Among sense-0-to-sense-1 items, 125/256 initially primed observations crossed the boundary (48.8%). The corresponding count for sense-1-to-sense-0 items was 109/324 (33.6%).

Several patterns recurred across analyses. *Spring* had the smallest H5 update, the lowest decoder endpoint adequacy in H4, and one of the lowest H5 transition rates. *Crane* had strong H4 recovery and the largest H5 conflict cost. *Bank* combined

a strong H0 baseline asymmetry with the largest H5 update.

The conflict endpoint was 0.140 below the matched control on average. The difference was -0.098 for encoders and -0.182 for decoders, and it was negative in 55 of 56 model-word cells. The conflict endpoint was also 0.056 below the resolver-only baseline. This difference was -0.018 for encoders and -0.093 for decoders, and it was negative in 45 of 56 cells (Figure 8).

Endpoint counts depended on the margin threshold. At zero, 404/784 states (51.5%) were on the resolved side. At $|\hat{M}| > 0.1$, 244 states (31.1%) were resolved, 247 (31.5%) were primed, and 293 (37.4%) were in the boundary region. At $|\hat{M}| > 0.3$, 61 states (7.8%) were resolved, 57 (7.3%) were primed, and 666 (84.9%) were in the boundary region. The continuous paired change provides the primary H5 summary.

G. Exploratory relationships across hypotheses

Absolute H0 carrier lean had little association with H1 selected-layer adequacy across the 56 model-word cells (Spearman $\rho = -0.143$).

For H5, carrier lean was aligned with the primed direction. Stronger primed alignment predicted a lower correct-sense margin at the homonym ($\rho = -0.227$; 112 cells) and resolver ($\rho = -0.251$; 112 cells). Its association with resolver-minus-homonym change was near zero ($\rho = -0.026$). Its association with transition rate was weak ($\rho = -0.146$; 107 cells).

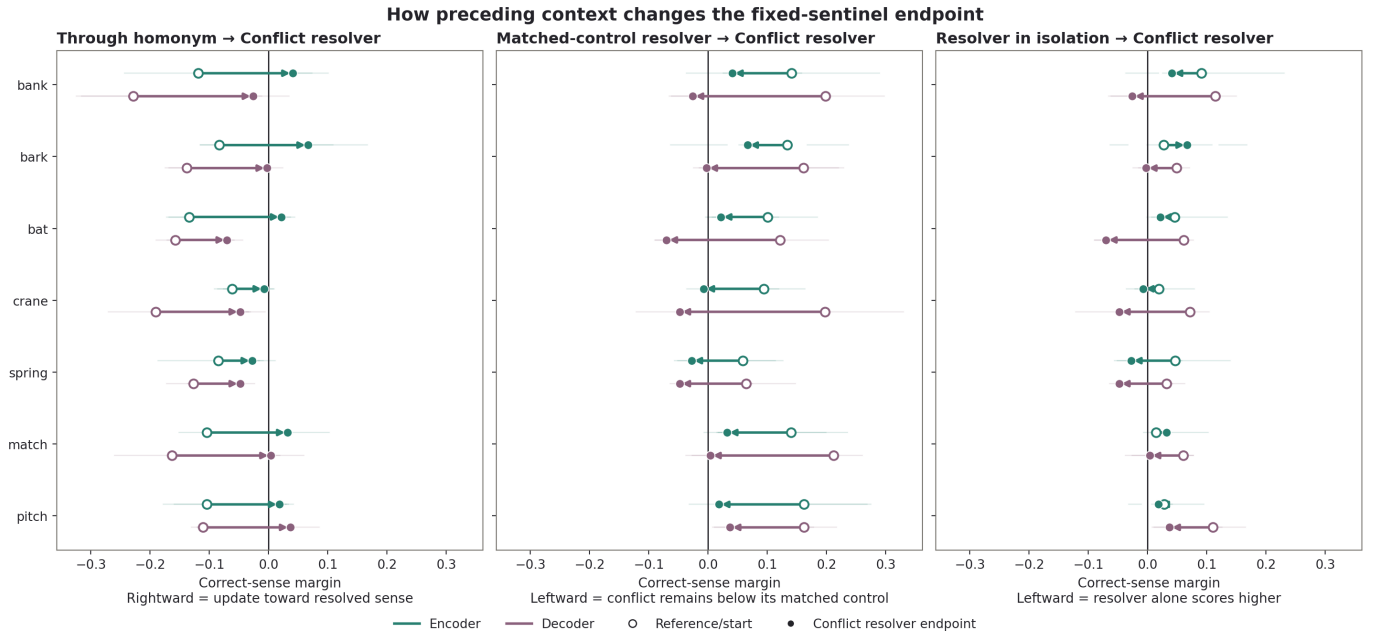


Fig. 8. Three H5 comparisons at the fixed sentinel. The left panel shows resolver updating. The middle panel shows conflict cost relative to the matched control. The right panel compares the conflict endpoint with the resolver-only baseline.

This pattern suggests a persistent starting offset. Stronger H0 primed alignment shifted both the homonym and resolver positions toward the primed side, while the resolver-induced movement stayed similar. *Bank* combined a strong baseline lean with the largest update. *Spring* combined a strong lean with the smallest update. These post-hoc patterns motivate confirmatory hierarchical analysis.

IV. DISCUSSION

Contextual sense was widely accessible across the sampled models. Its strongest readout varied across depth. Selected layers also varied across models, homonyms, and geometric criteria. Sense accessibility also depended on token position, available context, and context order.

A. Semantic localisation depends on criterion and stimulus

H0 showed systematic asymmetry in the two-sense geometry before explicit disambiguation. Some directions matched an intuitive dominant sense, such as the financial lean of *bank*. Other directions, including tree covering for *bark* and water source for *spring*, were less obvious. Bare-word and carrier directions also differed in half of the model-word cells. The baseline therefore reflects the combined influence of training distribution, carrier wording, tokenisation, target position, category breadth, and cluster shape.

These starting positions carried into H5. Stronger alignment with the primed sense predicted more primed homonym and resolver states. It had little relation to resolver-induced movement. The models therefore began from unequal positions while showing similar update magnitudes. This post-hoc pattern motivates confirmatory testing.

H1 and H2 also showed variable semantic depth. Intermediate layers improved held-out adequacy for some models, especially the sampled decoders. The mean advantage varied across model-word cells. GDV-based and adequacy-based selection often chose different layers. A stable semantic stage would require greater agreement across models, homonyms, and readout criteria.

The H2 results also clarify the source of the GDV disagreement. GDV summarises average class geometry. Adequacy tracks held-out decisions at the item level. A layer can improve average separation while leaving difficult sentences across the wrong centroid boundary. The Qwen2.5-7B *bat* example shows this pattern directly. Layer number alone therefore gives an incomplete account of semantic readout quality.

B. Context availability limits the layer-stage analogy

H3 showed that causal availability constrains homonym-position sense readout. Encoders used right context at the homonym. Causal decoders used only the preceding prefix. Additional decoder layers transformed that prefix without access to the later resolver. The resulting clause-order interaction follows from the attention masks and sets a clear limit on depth-based interpretation.

H4 showed that the missing decoder information became accessible at a later token. By the sentence-final period, the decoder had processed both the homonym and its resolver. Many previously inadequate cases then crossed into the correct local region. Sense information was therefore expressed at several positions within the completed sentence. Any claim about access to semantic information must specify both layer and token position.

Human comprehension also unfolds incrementally, but it combines recurrence, prediction, bottom-up input, and top-down context. Causal decoders use a fixed attention mask, while encoders process full context at once. H3 and H4 therefore compare access regimes. Their central result is that layer depth, reading time, and accumulated context form separate axes.

C. Semantic revision is path-dependent

H5 moved from accessibility to revision. The resolver shifted the fixed sentinel toward the authored sense across all models and homonyms. Movement was more common than boundary crossing. Many resolver states remained near the centroid boundary, and 40.3% of initially primed observations crossed into the resolved region.

The controls show a lasting effect of context order. Conflict items and matched controls shared the same homonym and resolver. Conflict endpoints still had lower correct-sense margins. The resolver-only condition also produced a strong sense signal. The preceding context shaped how strongly that signal appeared at the sentinel. Updating and persistence therefore occurred in the same readout.

Persistence supports continuity while earlier information remains valid. Sharp contradiction requires stronger revision. In H5, the resolver usually shifted the representation in the correct direction, while the earlier interpretation often retained influence. The balance between movement and boundary crossing indicates contextual inertia under the current geometric measure.

Human garden-path studies also report traces of an initial interpretation [10]. The present study provides a model-side measure of the same broad function: revision after conflicting context. Matched human data would allow direct comparison of initial interpretation, revision cost, and final understanding.

The sampled encoders reached higher resolver endpoints and smaller conflict-control differences than the sampled decoders. Decoder updates were sometimes larger because their homonym states began farther toward the primed sense. Model family, training data, parameter count, tokenisation, and depth also differed between groups. The architecture summaries therefore describe the sampled models as a whole.

D. Functional comparison across systems

Behavioural similarity motivates comparisons of computational function. A human and a transformer may both resolve *bank* from context, while their intermediate organisation differs. The present results challenge a direct mapping from transformer depth to a fixed sequence of human-like processing stages. Semantic readout varied across depth, metric, position, and context history.

Computational roles offer a clearer basis for comparison. Prediction, integration, conflict detection, and revision can be measured in both systems. The N400 is useful because it tracks semantic expectation and violation. The H5 conflict-control difference provides a model-side conflict measure for a future matched experiment. Trial-level comparisons could test whether model signals predict human behaviour or neural activity.

E. Limitations

The authored sense contexts vary in vocabulary, topic, syntax, target position, and length. These properties can influence the profiling geometry. Future stimulus sets should use counterbalanced templates, target masking, context-only classification, and lexically disjoint train-test splits.

The centroid axis covers two operational senses per homonym. Several words have additional meanings, and the two profiled regions may differ in breadth and variability. Adequacy and GDV therefore measure access within the tested binary contrast. Causal interventions, activation patching, and output measures would test whether the geometry contributes to model behaviour.

The H5 sentinel fixes token identity and terminal readout role across prefixes. Longer prefixes also change its absolute position. Length-matched filler controls, several sentinel tokens, and pooled sentence readouts would separate these effects and test generality.

The model sample is small and heterogeneous. Repeated observations within models, homonyms, and items also create dependence beyond the current summary analyses. Matched model families and hierarchical models would support stronger architecture claims. Human norms for ambiguity, prime strength, resolver clarity, reading cost, and final interpretation would support direct cross-system comparison.

F. Future work

A broader stage-order test should apply parallel held-out measures to surface form, syntax, contextual sense, task instruction, and output-relevant information. Counterbalanced contrasts could test whether these signals peak in a stable order, overlap across depth, or reorganise across model families. Layer studies should also compare broader depth regions.

The relation between H0 and H5 should be tested with a hierarchical model of baseline lean, prime direction, resolver strength, and update magnitude. Stronger stimulus controls would separate starting bias from item wording. H5 could also cover corrected entity attributes, changed event states, belief revision, and repeated updates over longer contexts.

A human experiment should use the same items to measure initial interpretation, reading or listening cost, final comprehension, and lingering misinterpretation. Trial-level correspondence would show how closely the model conflict measure tracks human revision.

G. Conclusion

The study found accessible sense structure across models and layers. Its location varied with the homonym, model, criterion, token position, and context history. Semantic-stage estimates therefore required more than layer depth.

Resolving context usually shifted representations toward the authored sense. Earlier conflicting context still reduced the final correct-sense alignment, and complete geometric resolution occurred in a minority of initially primed cases. Semantic readout in these models is distributed, context-sensitive, and path-dependent.

The results replace the search for one semantic layer with a more precise question: how semantic information becomes accessible, conflicted, and revised across computational systems.

REFERENCES

- [1] M. Schrimpf, J. Kubilius, H. Hong, N. J. Majaj, R. Rajalingham, E. B. Issa, K. Kar, P. Bashivan, J. Prescott-Roy, F. Geiger, K. Schmidt, D. L. K. Yamins, and J. J. DiCarlo, “Brain-Score: Which artificial neural network for object recognition is most brain-like?” *bioRxiv*, p. 407007, 2020.
- [2] R. Ryskin and M. S. Nieuwland, “Prediction during language comprehension: What is next?” *Trends in Cognitive Sciences*, vol. 27, no. 11, pp. 1032–1052, 2023.
- [3] I. Tenney, D. Das, and E. Pavlick, “BERT rediscovers the classical NLP pipeline,” in *Proceedings of ACL*, 2019, pp. 4593–4601.
- [4] N. Elhage, T. Hume, C. Olsson, N. Schiefer, T. Henighan, S. Kravec, Z. Hatfield-Dodds, R. Lasenby, D. Drain, C. Chen, R. Grosse, S. McCandlish, J. Kaplan, D. Amodei, M. Wattenberg, and C. Olah, “Toy models of superposition,” *Transformer Circuits Thread*, 2022. [Online]. Available: https://transformer-circuits.pub/2022/toy_model/index.html
- [5] T. Bricken, A. Templeton, J. Batson, B. Chen, A. Jermyn, T. Conerly, N. Turner, C. Anil, C. Denison, A. Askell, R. Lasenby, Y. Wu, S. Kravec, N. Schiefer, T. Maxwell, N. Joseph, Z. Hatfield-Dodds, A. Tamkin, K. Nguyen, B. McLean, J. E. Burke, T. Hume, S. Carter, T. Henighan, and C. Olah, “Towards monosemanticity: Decomposing language models with dictionary learning,” *Transformer Circuits Thread*, 2023. [Online]. Available: <https://transformer-circuits.pub/2023/monosemantic-features/index.html>
- [6] A. Templeton, T. Conerly, J. Marcus, J. Lindsey, T. Bricken, B. Chen, A. Pearce, C. Citro, E. Ameisen, A. Jones, H. Cunningham, N. L. Turner, C. McDougall, M. MacDiarmid, C. D. Freeman, T. R. Sumers, E. Rees, J. Batson, A. Jermyn, S. Carter, C. Olah, and T. Henighan, “Scaling monosemanticity: Extracting interpretable features from Claude 3 Sonnet,” *Transformer Circuits Thread*, 2024. [Online]. Available: <https://transformer-circuits.pub/2024/scaling-monosemanticity/index.html>
- [7] A. Schilling, A. Maier, R. Gerum, C. Metzner, and P. Krauss, “Quantifying the separability of data classes in neural networks,” *Neural Networks*, vol. 139, pp. 278–293, 2021.
- [8] J. Hewitt and P. Liang, “Designing and interpreting probes with control tasks,” in *Proceedings of EMNLP-IJCNLP*, 2019.
- [9] L. Frazier and K. Rayner, “Making and correcting errors during sentence comprehension,” *Cognitive Psychology*, vol. 14, pp. 178–210, 1982.
- [10] K. Christianson, A. Hollingworth, J. F. Halliwell, and F. Ferreira, “Thematic roles assigned along the garden path linger,” *Cognitive Psychology*, vol. 42, no. 4, pp. 368–407, 2001.
- [11] N. Kölbl, K. Tziridis, A. Maier, T. Kinfe, R. Chavarriaga, A. Schilling, and P. Krauss, “The predictive brain: Neural correlates of word expectancy align with large language model prediction probabilities,” *NeuroImage*, vol. 334, p. 121966, 2026.